

Tactical Missile Design - Chapters 1 & 2 Summary

CHAPTER 1: INTRODUCTION & KEY DESIGN DRIVERS

Missile Characteristics vs. Fighter Aircraft

Where missiles excel:

- Lateral acceleration: 30g+ (vs. aircraft ~9g)
- Longitudinal acceleration: 30g+ with developmental 400g+
- Dynamic pressure: Higher than aircraft
- Cost per unit: \$10K-20K vs. tens of millions (e.g., GBU-31 JDAM)
- RCS achievable: Comparable to low-observable aircraft without cockpit/sensor constraints

Where aircraft excel:

- Range (700+ nm for missiles vs. greater for aircraft)
- Kills per sortie: Aircraft carry multiple weapons
- Target acquisition: Human pilot superior to autonomous systems

Key Configuration Sizing Parameters

1. **Flight conditions:** α , M, h
2. **Nose fineness** - affects drag and seeker performance
3. **Diameter** - affects drag, seeker, warhead, subsystem packaging
4. **Propellant/fuel type & weight** - drives range, velocity
5. **Wing geometry/size** - maneuverability, efficiency
6. **Stabilizer geometry/size** - static margin
7. **Flight control geometry/size** - max angle of attack
8. **Length** - launch platform compatibility
9. **Thrust profile** - velocity time history

Key Equations from Chapter 1

Radar seeker detection range (two-way, active):
$$R_D = \left\{ \frac{\sigma P_t \lambda^2}{64\pi^3 K T B F L (S/N)} \right\}^{1/4}$$

where: σ = target RCS (m^2), P_t = transmitted power (W), λ = wavelength (m), K = Boltzmann's constant, T = receiver temp (K), B = bandwidth (Hz), F = noise factor, L = transmitter loss, S/N = signal-to-noise ratio

Radar seeker 3-dB beamwidth: $\theta_{3dB} = 1.02\lambda/d$

where: d = antenna diameter

****IR seeker detection range (one-way, passive):**** $R_D = \left\{ \frac{(I_\gamma)_{avg} \tau_a A_o D^* [\Delta f / (A_f f_{IFOV}^2 A_d)]}{(S/N)_d} \right\}^{1/2}$

where: $(I_\gamma)_{avg}$ = target radiant intensity (W/sr), τ_a = atmospheric transmission, A_o = optics aperture area (m²), D^* = specific detectivity (cm Hz^{1/2}/W), Δf = pixel bandwidth, A_f = focal plane area, f_{IFOV} = instantaneous field-of-view, A_d = detector area, $(S/N)_d$ = detection S/N ratio

Instantaneous field-of-view (IFOV): $IFOV = \frac{d_p}{(f\text{-number}) \times d_p} = \frac{d_p}{[d_o / (2.44\lambda)] \times d_p}$

where: d_o = optics aperture diameter, d_p = pixel diameter

Missile fineness ratio: $5 < \frac{l}{d} < 25$

Drag force: $D = C_D q S_{ref} = C_D q (\pi/4) d^2 = 0.785 C_D q d^2$

where: q = dynamic pressure (psf), S_{ref} = reference area (body cross-section)

Body first-mode bending frequency: $\omega_{BB} = 18.75 \left\{ \frac{Et}{W(l/d)} \right\}^{0.5}$

where: E = modulus of elasticity (psi), t = thickness (in), W = weight (lb), l/d = fineness ratio

Rule of thumb: $\omega_{BB} \geq 2 \times (\text{actuator bandwidth})$ for flight control stability

Correction of conceptual design prediction using baseline data: $P_{cp,c} = \frac{P_{s,c}}{P_{s,u}} \times P_{cp,u}$

where:

- subscript c = corrected (actual baseline data)
- subscript u = uncorrected (computational prediction)
- subscript s = baseline missile parameter
- subscript cp = conceptual design parameter

Spectral radiance (Planck's law) for IR target: $L_\lambda = \frac{3.74 \times 10^4}{\lambda^5 [e^{(4.4 \times 10^4)/(\lambda T)} - 1]} \text{ (W cm}^{-2} \text{ sr}^{-1} \text{ }\mu\text{m}^{-1})$

where: λ = wavelength (μm), T = target temperature (K)

Conceptual Design Process

Iterative steps:

1. Mission/scenario definition
2. Weapon requirements & trade studies
3. Launch platform integration constraints
4. Weapon concept design synthesis

5. Technology assessment & roadmap

Typical cycle: 3-9 months, 3-6 design iterations

Using Baseline Missiles

Benefits:

- Faster convergence
- Anchored with test data
- System engineering already balanced
- Establishes benchmarks

Limitations:

- Extrapolation $\pm 50\%$ with reasonable accuracy
- Don't get locked in—explore alternatives
- Project SOTA for electronics/sensors into future

Correction formula:

$$P_{cp,c} = \frac{P_{s,c}}{P_{s,u}} \times P_{cp,u}$$

where subscript c = corrected (actual), u = uncorrected (computed), s = baseline

CHAPTER 2: AERODYNAMIC CONSIDERATIONS

2.1 Missile Diameter Tradeoff

Fineness ratio: $5 < l/d < 25$

Small diameter advantages:

- Lower drag: $D = 0.785 C_D q d^2$
- Better launch platform compatibility

Large diameter advantages:

- Seeker detection range (proportional to d^4 for radar; d in aperture area)
- Warhead effectiveness (fragment velocity, blast, jet)

- Higher body bending frequency (reduces aeroelasticity)
- Better subsystem packaging
- Shorter length for platform compatibility

Example: Halving diameter $\rightarrow$ 1/4 drag; doubling diameter $\rightarrow$ 4 $\times$ drag

2.2 Nose Fineness Tradeoff

High fineness ($l_n/d = 5$):

- Aerodynamically superior
- Low observables
- Reduced propellant volume

Low fineness ($l_n/d = 0.5$ hemisphere):

- Electromagnetically ideal
- Maximum propellant length
- Low dome error slope

Compromise: $l_n/d \approx 2$ (tangent ogive, faceted, window, or multilens)

Supersonic wave drag:

$$(C_{D0})_{Body, Wave} = (1.59 + 1.83/M^2) \{ \tan^{-1} [0.5/(l_n/d)] \}^2$$

Bluntness effect: 10% nose tip bluntness adds negligible drag ($\sim 0.01 C_{D0}$)

2.3 Body Drag Components

Subsonic:

- **Skin friction dominant:** $(C_{D0})_{Friction} = 0.053(l/d)[M/(q)]^{0.2}$
- **Base drag:** $(C_{D0})_{Base} = 0.12 + 0.13M^2$

Supersonic:

- **Wave drag** dominates (function of nose fineness)
- **Base drag (coasting):** $(C_{D0})_{Base} = 0.25/M$
- **Base drag (powered):** Reduced by factor $(1 - A_e/S_{ref})$

2.4 Boattail

Best for subsonic missiles - drag reduction >50%

Supersonic: Limited benefit; large boattail angles cause flow separation

Rule of thumb: Boattail half-angle < 10°

2.5 Lifting Body vs. Axisymmetric Body

Normal force coefficient:

$$|C_N| = \left| \frac{a}{b} \cos^2 \phi + \frac{b}{a} \sin^2 \phi \right| \left[|\sin(2\alpha) \cos(\alpha/2)| + 2 \frac{l}{d} \sin^2 \alpha \right]$$

L/D advantage: Lifting body 40% higher at $q = 500$ psf, but comparable at $q > 5000$ psf

Disadvantages of tailored lifting body:

- Larger span (launch platform constraints)
- Inefficient solid subsystems packaging
- Requires advanced CFD/FEM analysis

2.6 Body Normal Force

Slender body (subsonic & supersonic):

$$(C_{N_\alpha})_{Body} = 2[(a/b) \cos^2 \phi + (b/a) \sin^2 \phi]$$

Independent of Mach number, function of angle of attack and body geometry only.

Body aerodynamic center location:

$$(x_{AC})_B/l_n = 0.63(1 - \sin^2 \alpha) + 0.5(l_B/l_n) \sin^2 \alpha$$

Moves from 63% nose length at $\alpha \rightarrow 0$ to 50% body length at $\alpha \rightarrow 90^\circ$

2.7 Static Stability & Control

Definition: Stable if $C_{m_\alpha} < 0$ (aerodynamic center aft of CG)

Static margin: $(x_{AC} - x_{CG})/d$

Rules of thumb:

- First mode body bending frequency $\geq 2 \times$ actuator bandwidth

- Control response ratio $|\Delta\alpha/\delta| > 1$ for adequate trim authority
- Relaxed static margin increases max trimmed angle of attack

Example: Neutral margin (SM = 0) vs. excessive margin:

- $\alpha_{\max}$ increases from 9.4° to 21.8° (+130%)
- $C_{N,\max}$ increases from 9 to 16 (+80%)

2.8 Flares vs. Tail Stabilizers

Flare advantages:

- Lower aerodynamic heating (critical for hypersonics)
- Smaller span requirement
- Less variation in stability with Mach

Tail advantages:

- Lower drag
- Available for flight control

2.9 Wings vs. No Wings

Wingless advantages:

- Better range (high supersonic)
- Better at high q and high angle of attack
- Better launch platform compatibility
- Lower RCS
- More propellant volume

Wing advantages:

- Better subsonic range and low q
- Higher maneuverability
- Lower guidance time constant
- Better seeker tracking
- Lower warhead wipe velocity

2.10 Surface Normal Force (Wing/Tail/Canard)

****High supersonic ($M > \{1 + [8/(\pi A)]^{0.5}\}$)** - Linear wing theory:**

$$(C_N)_{\text{surface}} = \frac{4|\sin \alpha' \cos \alpha'| \{(M^2 - 1)^{0.5}\} + 2\sin^2 \alpha'}{\frac{S_{\text{surface}}}{S_{\text{ref}}}}$$

****Low supersonic/subsonic** - Slender wing theory:**

$$(C_N)_{\text{surface}} = \left[\frac{\pi A}{2} |\sin \alpha' \cos \alpha'| + 2 \sin^2 \alpha' \right] \frac{S_{\text{surface}}}{S_{\text{ref}}}$$

where $\alpha' = \alpha + \delta$ (effective angle of attack)

2.11 Aerodynamic Center Variation

Subsonic: 25% mean aerodynamic chord

Supersonic: Shifts toward 50% at hypersonic

Prediction: $(x_{AC}/C_{mac})_{\text{surface}} = \frac{A(M^2 - 1)^{0.5} - 0.67}{2A(M^2 - 1)^{0.5} - 1}$ for $M > \sim 2$

2.12 Hinge Moment

$$HM = N_{\text{surface}}(x_{AC} - x_{HL})_{\text{surface}}$$

Increases with dynamic pressure and effective angle of attack.

Critical design note: Hinge must be at or forward of aerodynamic center ($x_{HL} \leq x_{AC}$) to avoid unstable feedback.

2.13 Surface Drag

****Skin friction dominant:****

$$(C_{D0})_{\text{friction}} = N_{\text{surface}} [0.0133 M / (q C_{mac})^{0.2}] (2 S_{\text{surface}} / S_{\text{ref}})$$

****Wave drag (Newtonian theory):****

$$(C_{D0})_{\text{wave}} = N_{\text{surface}} \frac{2}{\gamma M_A^2 R_T} \times \{...\} \sin^2 \delta_{LE} \cos \Lambda_{mac} b / S_{\text{ref}}$$

2.14 Surface Geometry Alternatives

Geometry	Advantages	Disadvantages
Delta	Low bending moment, low supersonic drag, high stiffness, low RCS	Large x_{AC} shift
Double-swept LE	Minimal x_{AC} shift, reduced HM, constant static margin	More complex geometry
Bow-tie	Low RCS, low span, low drag	Aeroelastic risk, larger bending moment
Trapezoidal	High control effectiveness, no major weaknesses	Moderate x_{AC} shift
Rectangular	Fits launch platform constraints	Large bending moment

2.15 Flight Control Alternatives

Two-axis (pitch/yaw or pitch/roll):

- Cheaper actuators
- Better packaging (e.g., Stinger)
- Disadvantage: Lower maneuverability

Four-axis (combined pitch/yaw/roll): Most common

Tail vs. Canard vs. Wing control:

- **Tail:** Efficient at high α , low HM, but decreased lift for statically stable
- **Canard:** Efficient packaging, increased lift, but stalls at high α (unless split)
- **Wing:** Small body rotation, fast response, but high HM, large induced roll, stall limited

Split canard advantage: Forward surface reduces local α on movable canard, enabling higher stall angle

2.16 TVC & Reaction Jet Control

Types:

1. Liquid injection: $\pm 7^\circ$ deflection
2. Hot gas injection: High deflection (valve temp $\sim 5500^\circ\text{F}$)
3. **Jet vanes:** $\pm 30^\circ+$ with 3-5% I_{sp} loss; share actuators
4. Axial plate: Coanda effect attachment

5. Jet tabs: Simple on/off
6. Movable nozzle: High cost but $\pm 30^\circ$ +
7. Reaction jets: Impulse or controlled duration (ms response time)

Advantages of unconventional control:

- Higher maneuverability at low q
- Faster response (reaction jets $<$ TVC)
- Reduced miss distance

2.17 Maneuver Laws

Skid-to-turn (STT):

- Direct command along line-of-sight
- Fast response
- Works with axisymmetric cruciform

Bank-to-turn (BTT):

- Roll first, then pitch
- Higher maneuverability for lifting bodies/wings
- Required for asymmetric inlets (minimize sideslip loss)
- Slower response; potential dome error slope issue

Rolling airframe:

- Bias roll rate (via thrust offset/tail incidence)
- Fewer gyros/actuators (low cost, small volume)
- Susceptible to roll resonance

2.18 Roll Orientation

Plus (+) orientation:

- Simpler control mechanization
- Lower trim drag
- Disadvantage: Statically unstable rolling moment ($C_{l\beta} > 0$)

X orientation:

- Better launch platform fit
- Higher L/D and static stability
- Higher pitch/yaw control effectiveness
- Statically stable rolling moment ($C_{l\beta} < 0$)

Conclusion: X preferred over +

2.19 Static Margin & Trim

Trim definition: $C_m = 0$

Static margin:

$$SM = \frac{x_{AC} - x_{CG}}{d}$$

Rule of thumb: $|\Delta\alpha/\delta| > 1$ for adequate deflection authority ($\sim 30^\circ$ limit)

Neutral stability trade: Reduces static margin enables:

- +130% increase in max α
- +80% increase in max trimmed C_N

2.20 Tail Area Sizing

****Neutral stability condition:****

$$\frac{S_t}{S_{ref}} = \frac{\{(C_{N_\alpha})_N[(x_{AC})_N - x_{CG}]/d + (C_{N_\alpha})_W[(x_{AC})_W - x_{CG}]/d\}(S_W/S_{ref})\} + (C_{N_\alpha})_B}{(C_{N_\alpha})_t[(x_{AC})_t - x_{CG}]/d}$$

Key insight: Required tail area increases with Mach (tail effectiveness decreases)

Wing location aft of CG reduces tail requirement (stabilizing); forward of CG requires larger tail.

2.21 Cross-Coupling Design Criteria

Keep cross-coupling moments < 30% of primary control derivatives:

- $|C_{l\delta_r}| < 0.3|C_{l\delta_a}|$ (roll due to rudder)
- $|C_{n\delta_a}| < 0.2|C_{n\delta_r}|$ (yaw due to aileron)
- $|C_{l\beta}| < 0.3|C_{l\delta_a}|$ (roll due to sideslip)
- $|C_{lp}| < 0.5|C_{l\delta_a}|$ (roll due to roll rate)
- $|C_m| < |C_{m\delta_e}|$ (pitch moment balance)

2.22 Body Buildup

****Total normal force (first-order):****

$$(C_N)_{total} \approx (C_N)_{body} + (C_N)_{wing} + (C_N)_{tail}$$

Assumes small interference effects; corrected by baseline missile aerodynamics.

Key Technologies Highlighted

1. **Faceted/window/multilens domes** - Low dome error slope + low drag
2. **Bank-to-turn maneuvering** - Enhanced maneuverability
3. **Lifting body airframes** - Improved L/D
4. **Double-swept surfaces** - Reduced HM, constant stability margin
5. **Neutral/relaxed static margin** - Increased maneuverability
6. **Lattice tails** - Higher control effectiveness, lower HM (but high RCS)
7. **Split canard control** - Enhanced high- α capability
8. **Free-to-roll tails** - Reduced induced roll, constant stability
9. **Unconventional control** - TVC, reaction jets for hit-to-kill capability